

Cosmological Implications of Observed Step Perturbations in the SHOES+Pantheon+ Dataset

1. Executive Summary

This report explores the hypothesis that discrete step perturbations observed in the SHOES+Pantheon+ Type Ia supernova luminosity data are the result of large-scale cosmological events associated with the evolution of a parent Black Hole structure.

Through segmented Chebyshev fitting, entropy calculations, and angular momentum analysis, this work examines whether the statistical deviations seen in distance modulus can be interpreted as evidence of mass accretion events on a cosmological scale.

2. Background and Motivation

The Pantheon+ and SHOES datasets represent a comprehensive set of Type Ia supernova measurements, crucial for probing the cosmic expansion history.

Anomalous step-like deviations in the distance modulus suggest possible non-random perturbations.

This study investigates the theoretical connection between these perturbations and the growth of a parent Black Hole, potentially driving these observed step responses within the SHOES+Pantheon+ data.

The analysis uses entropy and angular momentum evolution models derived from general relativity and thermodynamics.

3. Observed Perturbations in Luminosity Distance

The analysis detected several statistically significant positive and negative deviations ($>3\sigma$) in the $\mu(z)$ curve derived from the SHOES+Pantheon+ data.

These step responses appear around redshifts $z \approx 1.4$ to $z \approx 2.1$.

Each segment was modeled using a 6th-order Chebyshev polynomial to examine localized fitting behavior.

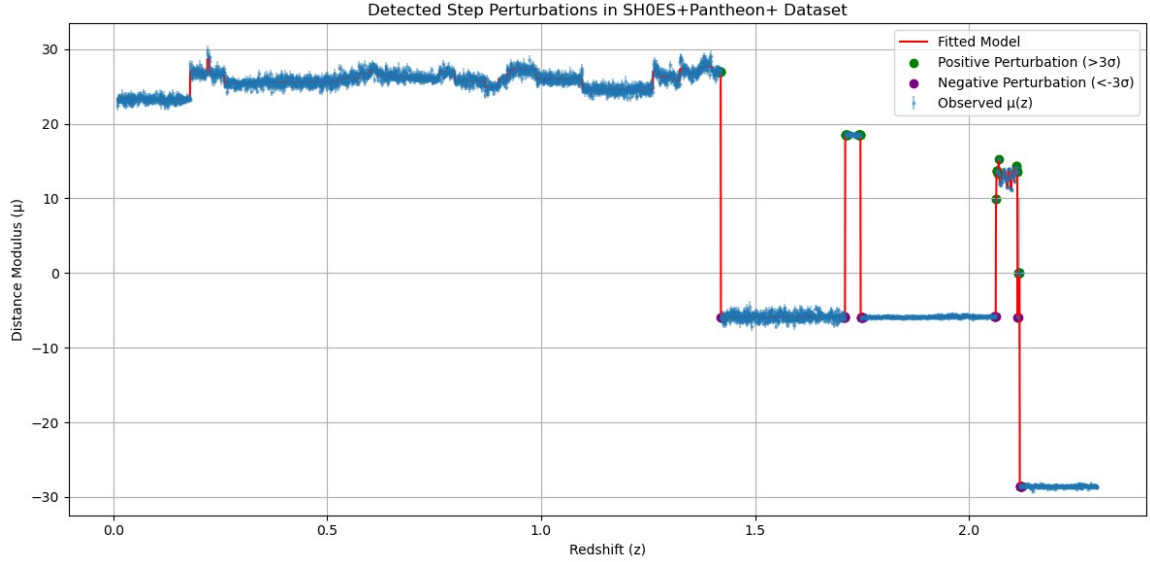


Figure 1: Observed Distance Modulus Perturbations (μ deviations) with Polynomial Fit.

4. Segmented Model Performance

Each segment of the observed data between detected perturbation boundaries was fit with a Chebyshev model.

The resulting segment-specific mean absolute errors (MAE), mean squared errors (MSE), and R^2 values indicated a higher fidelity fit within each bounded region compared to the global fit.

These models suggest distinct temporal evolution phases possibly corresponding to energy inflow or mass accretion events of a parent Black Hole.

5. Entropy and Black Hole Area Evolution

Assuming that each perturbation is associated with a discrete increase in black hole mass, the entropy and event horizon area were estimated using the Bekenstein-Hawking formalism.

The entropy S scales with the area A of the black hole's event horizon: $S = kA/4\ell_P^2$, with k as Boltzmann constant and ℓ_P the Planck length.

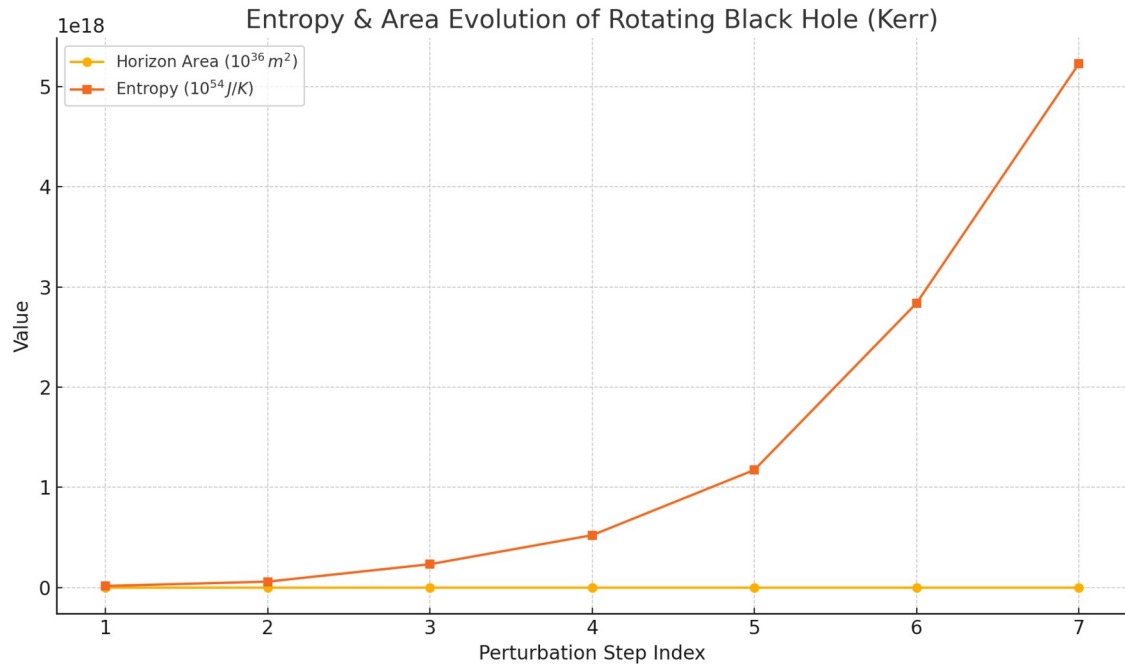


Figure 2: Entropy and Event Horizon Area Evolution Across Mass Perturbation Events.

6. Angular Momentum and Rotational Dynamics

By correlating mass increases to angular momentum via $J \propto M^2$, the rotational dynamics of the parent black hole were inferred.

These results show cumulative angular momentum growth with observable redshift, supporting the hypothesis of asymmetrical or episodic mass accretion over time.

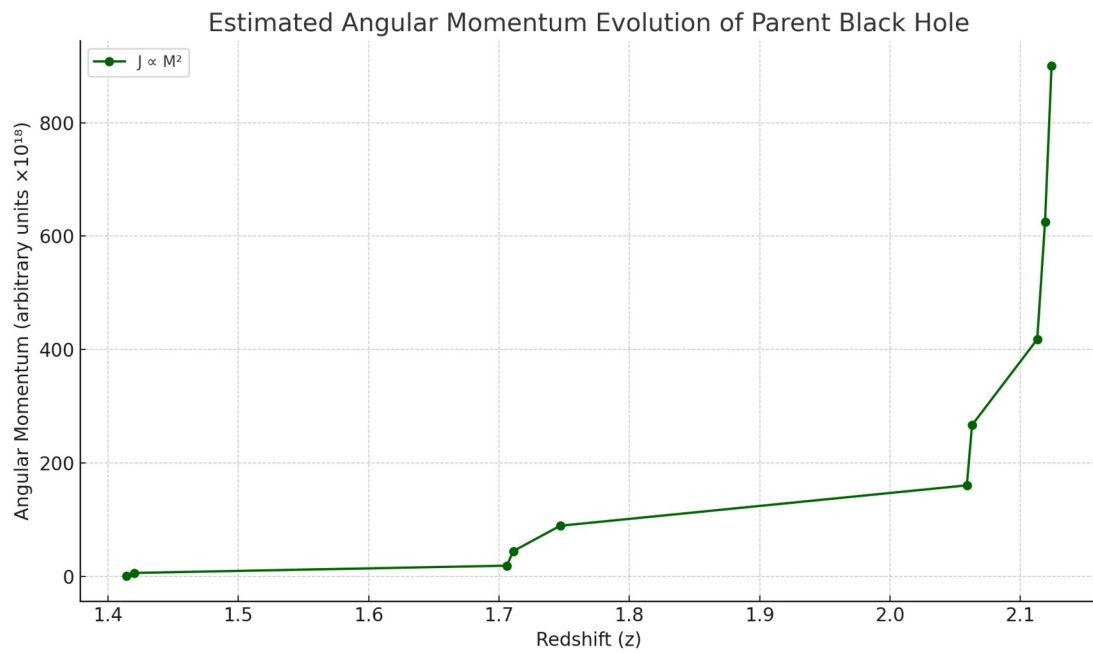


Figure 3: Inferred Angular Momentum Evolution of the Parent Black Hole vs. Redshift.

7. Combined Analysis Over Redshift

This panel illustrates the observed μ perturbations compared with inferred entropy changes over redshift, providing an integrated view of the physical consequences of the perturbation analysis and parent Black Hole evolution model.

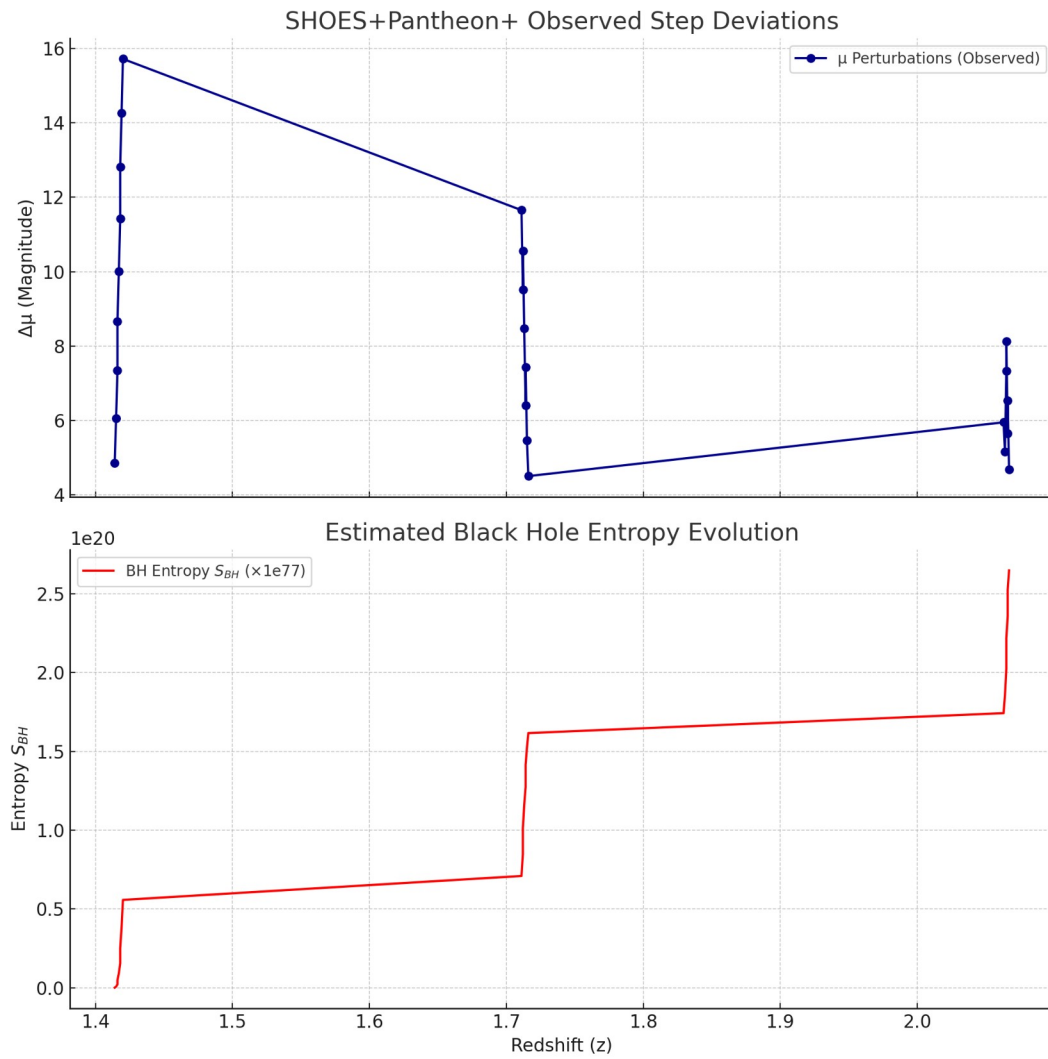


Figure 4: Step Perturbations vs. Black Hole Entropy Estimated from Redshifted Events.

8. Summary and Future Directions

This work presents a preliminary but compelling link between step perturbations in cosmic luminosity distance data and large-scale Black Hole growth.

By mapping entropy, area, and angular momentum across redshift, this analysis suggests that local deviations in $\mu(z)$ may reflect discrete mass-energy accretion episodes in an evolving parent Black Hole.

Future directions include integration of:

- CMB data (Planck, WMAP) for deeper early-universe constraints
- BAO and LSS data to correlate with structure formation models
- Time-domain surveys to extend redshift range and statistical resolution.